

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2018

Subject: 353152 Electronic Circuits I

Section: 5-6

Date: 3 October 2018

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 12 pages (including this page), 2 sections (37 questions) and a total score of 89 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators is allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (64 points) Multiple choices

Instruction: Mark the correct answer for the following questions in the provided answer sheet.

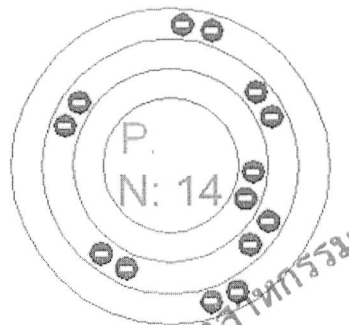
1. Which one of the following is semiconductor?

- (a) Ga (b) As (c) In (d) Ge

2. Valence shell contain up to _____ electrons.

- (a) 2 (b) 4 (c) 6 (d) 8

3. Atom of semiconductors contains 14 electrons. How many number of protons are there?

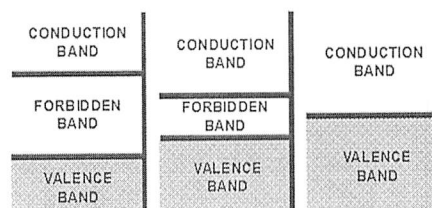


- (a) 14 protons (b) 4 positrons (c) 12 protons (d) 2 positrons

4. Which type of material has the narrowest energy gap between conduction band and valence band?

- (a) insulator (b) semiconductor
(c) conductor (d) resistor

5. Which type of material has the largest energy gap (Large forbidden band) between conduction band and valence band?



- (a) conductor (b) semiconductor (c) superconductor (d) insulator

6. Atoms of semiconductors contain _____ valence electrons.
(a) 2 (b) 4 (c) 6 (d) 8
7. Which one is the trivalent atom?
(a) Aluminum (Al) (b) Phosphorus (P)
(c) Bismuth (Bi) (d) Antimony (Sb)
8. How many of valence atoms of pentavalent atom or so-called N-type material?
(a) 3 (b) 4 (c) 5 (d) 6
9. Which one of following is true?
(a) Diode is open circuit when operating in forward bias mode.
(b) Forward bias means the voltage on p junction is lower than n junction.
(c) For a Germanium diode, there is no voltage drop for forward bias mode.
(d) The voltage drops of a Silicon diode is at 0.7 V when they are operating in forward bias mode.
10. Which circuit provides output as $V_{DC} = 0.636 (V_m - V_k)$?
(a) Half Wave Rectifier (b) Bridge Rectifier
(c) Full-Wave Rectifier (d) Center-Tapped Transformer Rectifier
11. Which one of following is true?
(a) Bridge-rectifier has lower voltage drop at diode compare to center-tapped transformer rectifier.
(b) The cost of bridge-rectifier equals to the center-tapped transformer rectifier.
(c) Half-wave rectifier provides higher dc voltage output than full-wave rectifier.
(d) Bridge-rectifier contains same number of diodes over center-tapped transformer rectifier.
12. Which of following transistor operating mode represents the open circuit?
(a) Cut-off (b) Saturation (c) Active (d) Reverse

13. Which of the following is true?

- (a) BE Junction is reverse biased when transistor is in active mode.
- (b) BE Junction is forward biased when transistor is in cutoff mode.
- (c) CB Junction is forward biased when transistor is in saturation mode.
- (d) CB Junction is forward biased when transistor is in active mode.

14. How much voltage drop over silicon diode when it is in forward bias mode?

- (a) 0.2 V
- (b) 0.3 V
- (c) 0.6 V
- (d) 0.7 V

15. What is the possible voltage drop over 2 silicon diodes in series when it is in forward bias mode from the source of 1 V?

- (a) 0 V
- (b) 0.7 V
- (c) 1.4 V
- (d) 1 V

16. An open circuit can have any voltage across its terminals, but the current is always ____.

- (a) 5 A
- (b) 0 A
- (c) 1 A
- (d) ∞

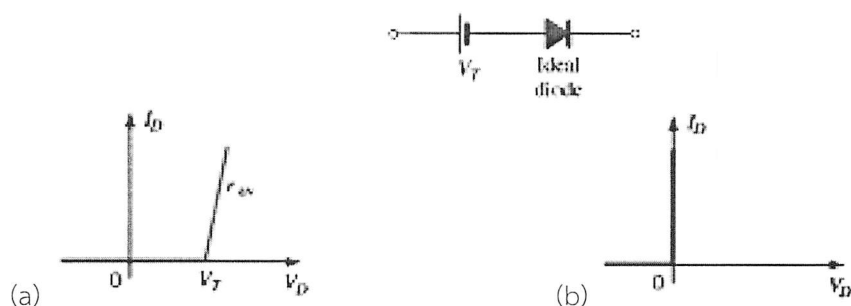
17. The intersection of the load line with the characteristic curve determines the ____ of the system.

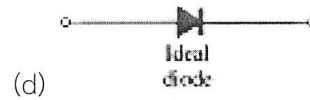
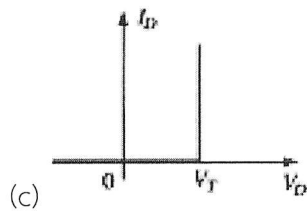
- (a) point of operation
- (b) load-line analysis
- (c) characteristic curve
- (d) forward bias

18. What is the output voltage of a voltage doubler circuit?

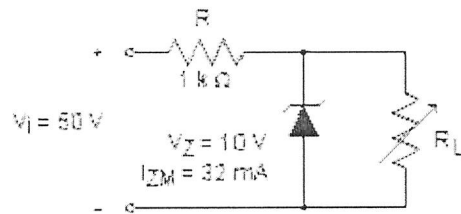
- (a) V_m
- (b) $2V_m$
- (c) $0.5V_m$
- (d) $0.25V_m$

19. Which one of the answers is corresponded to following figure?



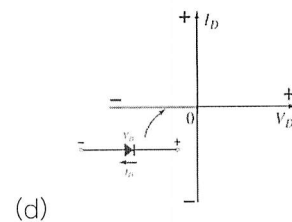
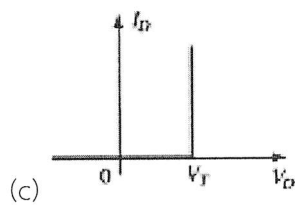
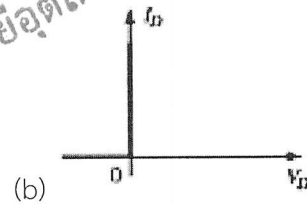
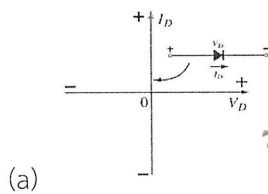


20. With this Zener diode in its "on state," what is the level of I_Z for the maximum load resistance?

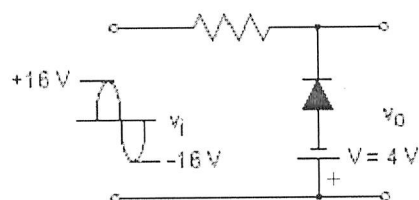


- (a) 0 mA (b) Undefined (c) Equal to I_{RL} (d) I_{ZM}

21. The ideal diode acts as an open switch. Which one is the diode characteristic?.

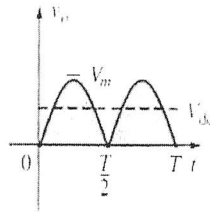


22. Determine the peak for both half cycles of the output waveform. Assume that the diode is ideal.



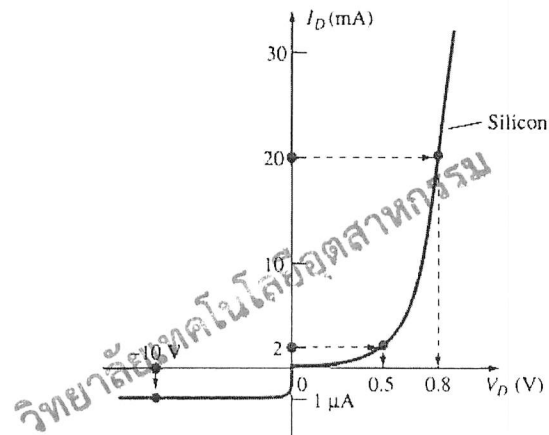
- (a) 16 V, -4 V (b) 16 V, 4 V (c) -16 V, 4 V (d) -16 V, -4 V

23. Which of the following is the average output value of full-wave rectifier?



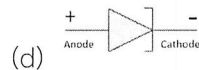
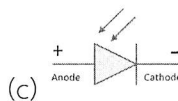
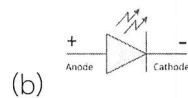
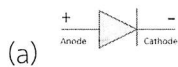
- (a) $0.318V_m$ (b) $0.5V_m$ (c) $0.636V_m$ (d) $1.414V_m$

24. From figure below, what is the DC resistance at 20 mA?

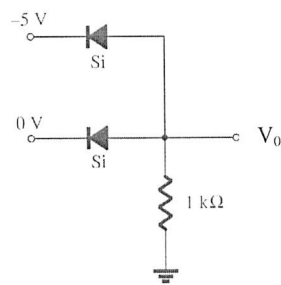


- (a) $4 \text{ m}\Omega$ (b) 4Ω (c) 40Ω (d) $40 \text{ k}\Omega$

25. Which one of the answers is the symbol of light emitting diode?

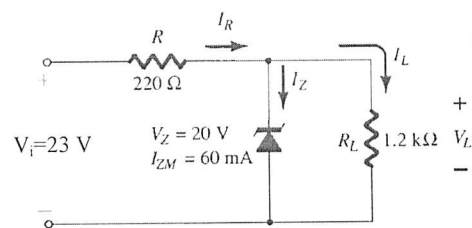


26. Determine V_0 from circuit below.



- (a) -5 V (b) -4.3 V (c) 0.7 V (d) 1.4 V

27. Determine V_L .

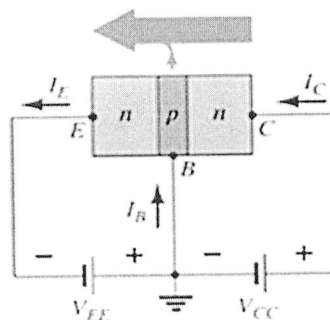


- (a) 23 V (b) 0 V (c) 3.5 V (d) 20 V

28. Determine I_Z .

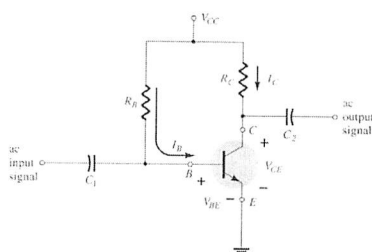
- (a) 0 A (b) 60 mA (c) 16.2 mA (d) 19.2 mA

29. What is a transistor configuration of the circuit in figure below?



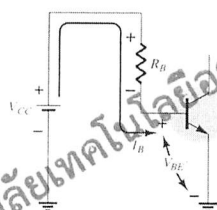
- (a) Common Emitter (b) Common Collector
(c) Common Base (d) Common Emitter feedback

30. What is a bias type of the circuit in figure below?



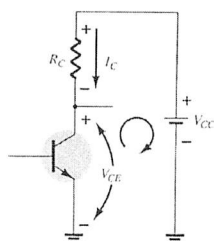
- (a) Fixed-bias circuit
(b) Emitter-stabilized bias circuit
(c) Voltage divider bias circuit
(d) DC bias with voltage feedback

31. From Base-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?



- (a) $V_{CC} - I_B R_B - V_{BE} = 0$
(b) $V_{CC} - I_B R_B + V_{BE} = 0$
(c) $V_{CC} + I_B R_B - V_{BE} = 0$
(d) $V_{CC} + I_B R_B + V_{BE} = 10$

32. From Collector-Emitter loop in picture below, what is the Kirchhoff's voltage equation for the loop?

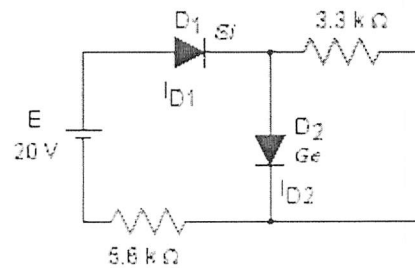


- (a) $V_{CE} + I_C R_C + V_{CC} = 0$
(b) $V_{CE} - I_C R_C + V_{CC} = 0$
(c) $V_{CC} - I_B R_B - V_{BE} = 0$
(d) $V_{CE} + I_C R_C - V_{CC} = 0$

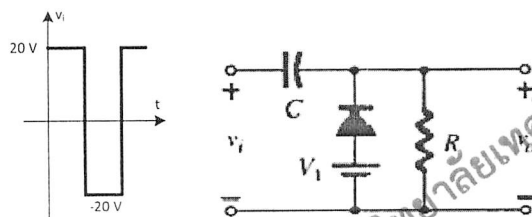
Part 2 (25 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

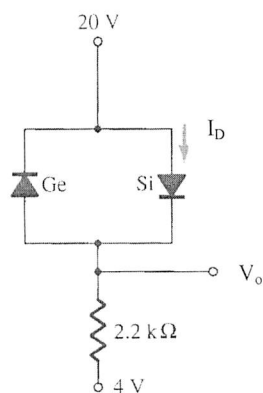
1. Determine I_{D1} from figure below



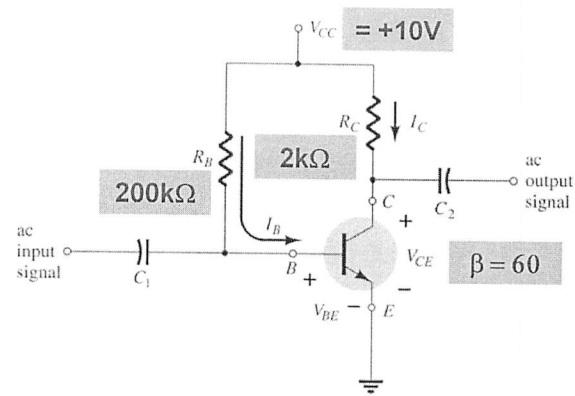
2. Sketch the output waveform of the network below. Assume that the diode is ideal.



3. From the circuit, determine V_o and I_D .

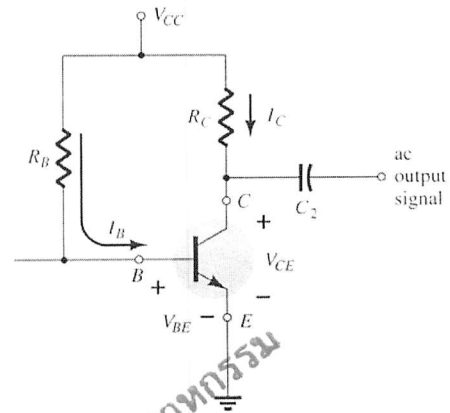
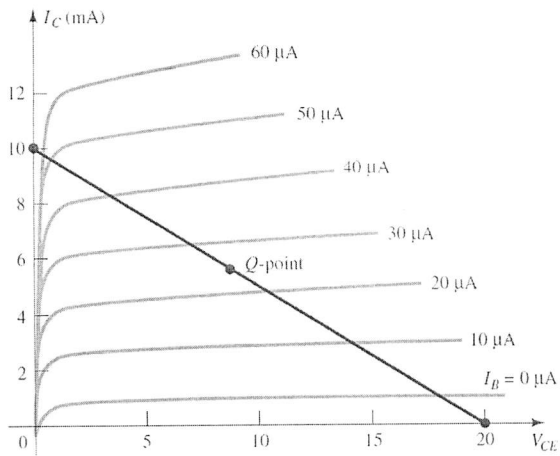


4. Determine I_B and I_C of the circuit below.



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5. Given the load line and defined Q point of figure below, determine the required values V_{CC} , R_C and R_B for a fixed-bias configuration.



Answer Sheet

Name: _____ ID: _____

Subject: 353152 Electronic Circuits I

	a	b	c	d
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